Project Proposal

Project: Supply Chain Performance Analysis: Shipment Delays, Cost Efficiency, Inventory Availability, and the Impact of Shortages on Patient Cases

Objective: Measure what the data actually supports: the delivery cycle and cost of the historical shipments in the shipments table (in), followed by the analysis of purchase orders, inventory and patient cases in the hospital layer, and convert the results into decisions supported by numbers that can be recalculated. The project pursues three objectives: (1) describe the shipment delivery cycle and cost: schedule status, lead time, customs clearance, shelf life at delivery, vendor value and freight cost; (2) measure supplier performance, stockout, unmet demand and wastage in the hospital layer (purchase orders and daily inventory); (3) describe patient case outcomes by treatment status in the hospital layer and their descriptive link to item stockouts, without causal claims.

Description: A five-page Power BI analysis of supply chain performance, delivered in the Data Analytics track. The analysis answers 21 analytical questions with 38 visuals; each question is tied to specific columns, a written KPI definition and a chart that the source data can support. The source is one workbook (Data_Final_project_IR.xlsx) with two separate data families:

- **Shipments (sheet in):** 46 data rows, 37 columns, 46 unique id values, 40 distinct po / so # values, 7 countries, 11 vendors and 3 product groups; the readable date columns span 7 November 2007 to 18 August 2015.
- **Hospital layer:** items (30 items: 14 Critical, 16 Essential), suppliers (9), hospitals (4), inventory_daily (10,793 rows), purchase_orders (2,187 rows) and patient_cases (5,665 rows); inventory dates run from 1 January to 31 March 2026 according to the data dictionary.
- **Reference sheets:** data_dictionary (59 definitions), schema_relationships (9 relationships), kpi_definitions (11 definitions) and issues_answer_key (107 documented issues).

Boundaries applied throughout the project:

- No figure is written from memory. Every reference number is calculated directly from the source file and recalculated after cleaning, with the effect of each change documented.
- All 46 shipments in sheet in were delivered on or before the scheduled date (44 on schedule, 2 earlier, 0 later) and shipment mode is Air in 45 rows. The shipment analysis is therefore built on schedule status, lead time, customs clearance, shelf life, vendor value and freight cost, not on delay rankings or mode comparisons.
- Shipments and hospital tables are separate families. There is no direct relationship or causal inference between them, and no historical shipment is linked to a patient or a hospital.

- The hospital layer is documented in the data file as fictional/synthetic. It is presented as an illustrative analytical case, not as medical evidence that a supplier delay caused a death or a complication.
- Blank values remain NULL and are never converted to zero. Outliers are verified, flagged and explained; they are not deleted because they are large.
- Small groups are not ranked: countries with fewer than 5 shipments and vendors with fewer than 3 shipments are grouped as Other, and every visual shows its sample size.

Tools & Design: Excel for manual inspection and documentation; SQL for cleaning, validation and repeatable calculations; Python (pandas and matplotlib) for recalculating the main KPIs and drawing basic descriptive charts; Power BI for the model, DAX and visuals; GitHub for version tracking; Word/PowerPoint or PDF for documentation and presentation.

- **Dashboard design:** five pages - Executive Overview; Delivery & Cycle Time; Suppliers & Cost; Purchase Orders & Inventory; Case Impact & Decision Quality - covering questions Q01 to Q21 with 38 visuals (KPI cards, column, bar, Pareto, histogram, scatter, box plot, heatmap, line, donut and tables).
- **Python scope:** one notebook that loads the cleaned dataset, recomputes the headline KPIs of each family, draws basic histograms with an outlier check (IQR rule), and compares the results with the SQL and Power BI figures.
- **Data flow:** source files, Raw layer, Working layer, SQL Staging, Power BI models (shipments and hospitals kept separate), dashboard pages.
- **Execution principle:** for every result - source and specific column, documented cleaning rule, written KPI definition, SQL/Python calculation, comparison with Power BI, and an explanation of the limits of the conclusion.

Technologies to Use: SQL; Python 3 (pandas, matplotlib); Power BI Desktop (Power Query, Model view, DAX, Report view); Microsoft Excel; GitHub (one repository with the folders sql, python, powerbi, data and docs; a main branch plus one branch per member; pull request before merge); Microsoft Word/PowerPoint or PDF.

Team Members & Responsibilities:

1- **Hossam - Technical Analytics Lead:** Project proposal; KPI specification and KPIs document; GitHub repository; SQL staging and integrity checks; Python notebook (KPI recalculation and basic charts); software architecture, DFD and technology stack; dashboard Pages 2 (Delivery & Cycle Time) and 5 (Case Impact & Decision Quality); master reconciliation; technical sign-off and final refresh QA.

2- **Yahia - Data Preparation & Testing Lead:** Raw and Working data layers with profiling; project plan and task assignment; cleaning of the shipments and hospital tables; derived fields and cleaned dataset; Power BI data models and core measures; dashboard Page 4 (Purchase Orders & Inventory); test plan, UAT and bug log; project infographic.

3- **Ola - Data Quality & Review Lead:** Stakeholder analysis, user stories and use cases; literature review file; data quality rules and mapping sign-off; analytical question map;

problem statement and use case diagram; dashboard Page 1 (Executive Overview); findings and recommendations; project presentation.

4- **Aya - Data Dictionary & Requirements Lead:** Data dictionary; functional and non-functional requirements; risk assessment; before/after reconciliation; wireframes and UI/UX guidelines; ER diagram and logical & physical schema; dashboard Page 3 (Suppliers & Cost); README and execution guide; user manual and final documentation.

Week 1 (11-15 October 2026): *Scope, Definitions, Raw/Working Layers and Planning Documents*

- **Hossam:** KPI Specification and KPIs Document; GitHub Repository and Branching Strategy; Project Proposal.
- **Yahia:** Raw and Working Data Layers with Data Profiling; Project Plan and Task Assignment.
- **Ola:** Stakeholder Analysis and User Stories & Use Cases; Literature Review File (Lecturer Input).
- **Aya:** Data Dictionary; Functional & Non-functional Requirements; Risk Assessment & Mitigation Plan.

Deliverables:

- KPI Specification + KPIs Document
- Repository + Branching Strategy
- Raw/Working Data Layers + Profiling Report
- Project Plan + Task Assignment & Roles
- Project Proposal
- Stakeholder Analysis + User Stories & Use Cases
- Literature Review
- Validated Data Dictionary
- Functional & Non-functional Requirements
- Risk Assessment & Mitigation Plan

Exit gate: Planning, literature review and requirements documents are ready for submission on 16 October 2026.

Week 2 (18-22 October 2026): *Data Cleaning, Integrity Checks and Quality Rules*

- **Hossam:** SQL Staging, Duplicate and Key Integrity Checks.
- **Yahia:** Shipments Table Cleaning; Hospital Tables Cleaning.
- **Ola:** Data Quality Rules and Mapping Sign-off.
- **Aya:** Before/After Reconciliation.

Deliverables:

- SQL Staging Layer + Integrity Report
- Clean Shipments Table + Cleaning Log
- Clean Hospital Tables + Cleaning Log
- Data Quality Rules + Categorical Mapping Sign-off
- Before/After Reconciliation Report

Exit gate: Cleaning logs and documented mappings are delivered; before/after differences are explained.

Week 3 (25-29 October 2026): *Derived Fields, Cleaned Dataset, Python Notebook and Dashboard Design*

- **Hossam:** Python Notebook: KPI Recalculation and Basic Charts.
- **Yahia:** Derived Fields and Cleaned Dataset.
- **Ola:** Question-to-KPI-to-Chart Map.
- **Aya:** Wireframes & UI/UX Guidelines.

Deliverables:

- Python Notebook (pandas, matplotlib)
- Derived Fields + Cleaned Dataset
- Analytical Question Map
- Dashboard UX Specification (Wireframes + Guidelines)

Exit gate: Derived fields are validated and the analytical question map is agreed.

Week 4 (1-5 November 2026): *System Analysis & Design Pack and Power BI Data Models*

- **Hossam:** Software Architecture, DFD and Technology Stack.
- **Yahia:** Power BI Data Models (Shipments and Hospitals, Kept Separate) and Core Measures.
- **Ola:** Problem Statement and Use Case Diagram.
- **Aya:** ER Diagram and Logical & Physical Schema.

Deliverables:

- Software Architecture + DFD + Technology Stack
- Validated Data Models + Core Measures
- Problem Statement + Use Case Diagram
- ER Diagram + Logical & Physical Schema

Exit gate: Design documents are ready for submission on 6 November 2026; relationships, denominators and filters are tested.

Week 5 (8-12 November 2026): *Dashboard Build: Five Pages*

- **Hossam:** Dashboard Page 2: Delivery & Cycle Time; Dashboard Page 5: Case Impact & Decision Quality.
- **Yahia:** Dashboard Page 4: Purchase Orders & Inventory.
- **Ola:** Dashboard Page 1: Executive Overview.
- **Aya:** Dashboard Page 3: Suppliers & Cost.

Deliverables:

- Dashboard Page 1
- Dashboard Page 2
- Dashboard Page 3
- Dashboard Page 4
- Dashboard Page 5

Exit gate: Every visual is linked to a specific question, column and KPI.

Week 6 (15-19 November 2026): *Reconciliation, Testing, UAT and Findings*

- **Hossam:** Master Reconciliation (SQL, Python, Power BI).
- **Yahia:** Test Plan, Functional Testing, UAT and Bug Log.
- **Ola:** Findings and Recommendations.
- **Aya:** README and Execution Guide.

Deliverables:

- Master Reconciliation Report
- Test Plan + Dashboard Functional Test + UAT & Bug Log
- Findings & Recommendations
- README + Execution Guide

Exit gate: No critical error remains open.

Week 7 (22-26 November 2026): *Sign-off, Infographic, Presentation and Final Documentation*

- **Hossam:** Technical Sign-off and Final Refresh QA.
- **Yahia:** Project Infographic (Single 16:9 Image).
- **Ola:** Project Presentation (PPT/PDF).
- **Aya:** User Manual and Final Documentation.

Deliverables:

- Technical Sign-off Package + Final Refresh QA
- Project Infographic

- Project Presentation
- User Manual + Final Documentation

Exit gate: Final internal delivery.

DEPI Submission Dates

- **16 October 2026:** Project Planning & Management, Literature Review and Requirements Gathering
- **6 November 2026:** System Analysis & Design
- **26 November 2026:** Final internal delivery
- **30 November 2026:** Implementation (Source Code & Execution)
- **4 December 2026:** Final Presentation, Testing & Reports
- All documents are uploaded to GitHub.